Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

loops, functions, conditionals, list, dataclasses

Improve literacy with matplotlib and possible numpy

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

easiest : implementing math and the different logical gates

Hardest: mapping the math onto a coordinate grid and drawing it on the website with matplotlib

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

Drawing the sphere on the website, should definitely start with implementing the

different gates

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Distractions could include diving to far into the mathematical implementations rather than abstracting them away.

1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

Help Page

Brief explanation paragraph

And some photos

Settings Page

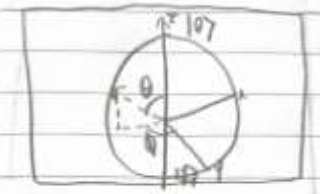
Settings

Color Drop down

☐ Show notation

☐ Show Others

Main Page



Quantum Gates

Settings

PX PY PZ

Select Box For Rgate

Text box for angle

H S T

Apply Rotation

History Page

Hadamard – State transformed into
Pauli – X – - -
Pauli – Y – - -
Hadamard – - -
Pauli – Z – - -
- - -

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**Education tool for college level students looking to get into quantum computing.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	gate_history: list[Gate] List of all applied gates to be able to view later on updates accordingly when gate applied X_coordinate: float x value for the vector updates accordingly when gate applied Y_coordinate: float y value for the vector updates accordingly when gate applied z_coordinate : float z value for the vector updates accordingly when gate applied background_color: str - string holding the color of the background Sphere_shown: bool - whether or not the sphere will be drawn show_notation : bool - whether or not the notation is shown
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified	Gate name: str name of the gate description: str what the gate does gate_type: str mathematical family of the gate

(if they are modified)	axis: str axis of the plane that the gate relates to Will not be modified, this will be used to avoid hard coding values into gate description and instead store them in dataclass instances and to more easily use them and add to a list
Routes (minimum 7) The names of your route functions	index Simulator_page pauli_X_gate pauli_Y_gate pauli_z_gate H_gate S_gate T_gate Rotation_gate settings gate_history undo_gate
Pages (minimum 5) A brief description of the pages in your site	Help page - explains the different gates Simulator page - page displaying the bloch sphere and the gates Gate history page - displays all gates that have been applied to the qubit Settings page - allow users to configure page layout **Professor said 4 was acceptable with sufficient complexity elsewhere**
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes	text box for input of angle of rotation, Select box to determine which rotation gate(Rx, Ry, Rz), Check boxes on preferences page to alter where things are placed, the color of the page, show notation

For more information on these input types, see Drafter Documentation.	
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	Based on input of the select box perform a different rotation If statement used to determine whether or not the sphere is shown If statement used to determine whether or not the notation is shown
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	Loop through list of gate history and print them on the history page Used to view the history of the applied gates

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Replace this text with your answer.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.